

Element A

A1: The janitors need a solution to keep the trash bags from falling into the trash can. It causes trash to pile up inside the bin, but outside the bag, and also causes it to fall out of the trash bin. This makes them have to do more work, having to reach for the extra trash and pick it up, which is unsanitary. Not to mention the mess and smells that it creates.

A2: A trash bag falling out of a trash can is a common problem occurring in households and public places every day. The reason is that the trash bag slips off the corners of the trash can and falls down into the trash. The trash bags aren't strong enough to hold onto the corners of the trashcans and slip down due to this. The weight of the trash in the trash bags pulls down and causes tension on the trash can. Without a good enough hold or grip the bag will slip down. In the problem the people who own the trashcan are involved, they put the trash bag into the trashcan and attach it to the edges of the can. This happens in households or public locations. This happens pretty frequently and is a pretty big issue for many people.

A3: This problem involves two main stakeholders. The first stakeholder are the janitors. They are involved because the problem affects their job, by making them stop to clean up the mess. Solving this problem would make their jobs easier and make them not have to stop and fix the trash can. The second stakeholder who is involved in the problem are the students at the school. This problem affects the students because it causes a mess in the cafeteria and classrooms that they have to deal with. The trash bag falling can lead to an unsanitary workspace. Solving this problem would overall improve the quality of life that students have in school.

A4: This problem's primary stakeholders are the janitors since they have to deal with the problem of the trash bags falling down and they have to clean up the mess. The secondary stakeholders are the students and teachers because they might not be able to throw their trash away because the trash bag fell down into the trash can, this can make the school very unsanitary.

A5: This problem happens when the janitors are cleaning the cafeteria, when students are throwing away their trash, when people are cleaning up classrooms, your kitchen or when you're decluttering and throwing things away. When things pile up in the trash can it starts to weigh the bag down a lot. And when doing things like this, that can cause the bag to fall off the rim of the trash can very often. This happens daily because these situations occur everyday.

A6: It's important to solve the problem with accurate measurable data to make sure that the product works as intended. Things like how big the device will be and how long it's meant to last affect the efficiency of the product.

Element B

B1: This problem has had prior attempts. Tying a knot around the trash bag to keep the trash bag tight has been attempted. Although this attempt has worked, many problems can and have arisen from it. The knot can easily come undone and make for the trash bag to become loose. The knot is a cheap and simple way of fixing the problem that does not require outside elements but does come with some flaws that make for an outside element or solution to work better.

B2: Previous attempts to solve this problem included tying a knot around the trash can which proved to work well. However, this solution is not permanent and cannot be used every time to fully solve the problem.

Element C

C1:

- 1.) Rim fits tightly inside of clip
- 2.) Max height of 1 $\frac{3}{4}$ in
- 3.) Max width of 1 in
- 4.) Max length of 1 $\frac{1}{4}$ in
- 5.) At least 4 clips around the rim

C2:

- 1.) Rim fits tightly inside of clip (1)
- 2.) At least 4 clips around the rim (2)
- 3.) Max width of 1 in (3)
- 4.) Max length of 1 $\frac{1}{4}$ in (3)
- 5.) Max height of 1 $\frac{3}{4}$ in (3)

C3: 2 of the design requirements are listed as our high priorities. These include the rim fitting tightly inside of the clip and having at least 4 clips around the rim.

C4: All requirements/parameters are objective and measurable.

C5: The design of the clips requires the trash bag to stay up for extended periods of time and to stop the amount of trash that falls into the bottom of the bin throughout the day. The clips are meant to be small so that they can be attached and removed from the rim of the trashcan easily.

C6: Our design requirements for our trash bag solution, our clip, meets what our primary stakeholders, the janitors, needs. We explained our ideas to the janitor and they said it was a great solution. From the questions we asked them, they have tried big binder clips to keep it up but ultimately had trouble with it. They told us how hard it is to clean up something when the

trash bag falls in the trash can. So from the information they told us, we worked our design around their feedback and made our solution to meet their needs.

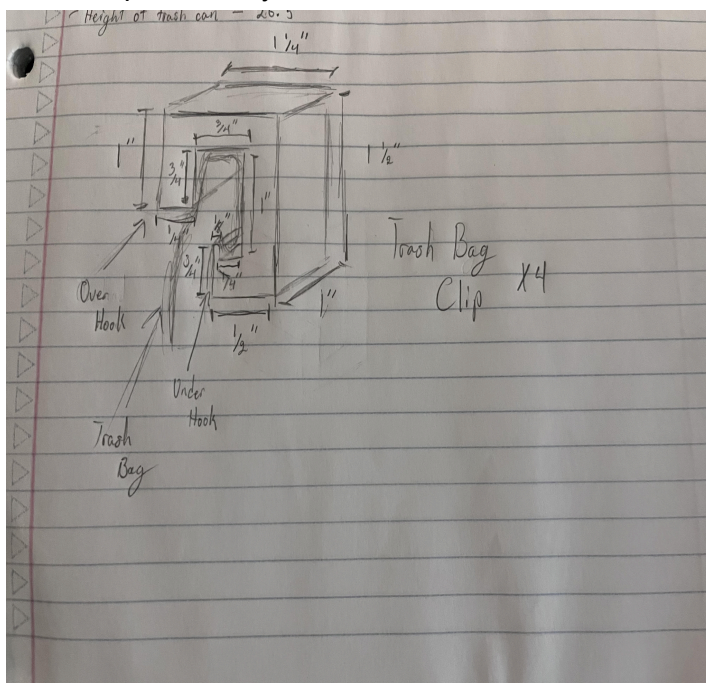
Element D

D1: When we created our solution, we compared it to possible solutions and went through a process to do this. For this, we looked on the internet and found some other solutions people came up with for the problem. We found many different solutions and from these, we found the flaws and pros of each solution and worked our design to try and find the best solution for this problem.

D2: The design we chose for the solution was small bag clips that are meant to prevent the trash bag from falling into the bin,

D3: To take a plan of action to build our design, we first drew a rough sketch of what we had in mind for our design. For this sketch we made sure to add details like labels and measurements so when we went to build our prototype and our actual design, we had a reference. Our plan for after the drawing is to design, we are going to create our 3D model prototype so we can see what it will look like when it comes to life (roughly). After our prototype we plan to 3D print our design. This will be our final step in our design process. Through the sketch, prototype and our final product, we will make minor changes to our design to accommodate things we think would be better and our stakeholders wants and needs for this design solution.

D4: The attributes of the design our group chose are that we will 3D design clips that will keep the trash bag in its place and tightly hold together the bag onto the trash can. There will be at least 4 clips and they will have a max width of 1, max length of $1\frac{1}{4}$, and max height of $1\frac{3}{4}$.



Element E

E1: Our proposed solution is supported by STEM principles by showing the drawings and sketches our team created to lead to the final design of the product. By researching other people's solutions to similar problems, we were able to come up with an effective design that's able to support and hold up the trash bag.

E2: Our proposed design solution uses stem in a couple ways. Science is used in the design process, in determining whether or not the solution will hold the bag up sufficiently. Technology is used to create the solution, as we need to 3D print it. Engineering and Math are used hand-in-hand to have the correct measurements for the size of the design solution so that it will not only keep the bag from falling, but also so that it will fit properly on the trash can.

E3: Our expert Mr. Thib says our project represents stem. He was mainly concerned with how the size of the trash can would affect the size of the clips needed. "It all just depends on what size the trash bags are because all kinds aren't the same and you just have to be careful because if it's mass produced, different cans have different size lids."

E4: The field experts that we have reviewed and collected our information, were not one but two opinions from Janitors, who are experts in cleaning and keeping the school clean and sanitary, which is what we are working on right now. We also interviewed a regular classroom teacher who is an expert with throwing out things in the classroom each day. Those are the field experts that we were working with.

E5: For the interview janitor Joe stated how people throw trash on top of the bag when it falls out and creates a mess. This problem occurred pretty often before using binder clips as a solution which decreased how often it happened. This takes 10-15 minutes depending on what was thrown in the trash. Coffee takes the longest to clean up. A solution they tried was tying a knot on the side of the bag (didn't work), and binder clips (which"worked pretty well). Also when interviewing Mr.Thibideau he stated that the solution should work if measured correctly. The clip must be wide enough and strong enough to hold onto the lip and secure the trash bag.